

Lecture 19 — Matrix-vector multiplication

Wed Oct 7

Last updated: August 18, 2026.

The class outline for this day will be posted before class.

Related reading and the course-wide list of daily objectives: [Lectures](#).

Learning objectives

By the end of class, students should be able to:

1. Compute Ax by hand for a small matrix.
2. Interpret Ax as a linear combination of the columns of A .
3. Interpret A as an input-output transformation.
4. Explain how the dimensions of A determine the input and output spaces.